

ARCH and GARCH Models

Volatility clustering · Conditional heteroskedasticity · MLE · VaR

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Learning objectives

By the end of this chapter, you will be able to:

- recognise the limits of homoskedastic ARMA on financial returns;
- state the ARCH(q) and GARCH(p, q) models, their stationarity conditions and unconditional moments;
- run Engle's LM test for ARCH effects on the residuals of a mean model;
- estimate a GARCH model by maximum likelihood and validate the standardised residuals;
- produce a one-step-ahead conditional volatility forecast and a Value-at-Risk estimate.

Chapter outline

1. Stylised facts of financial returns
2. Conditional vs unconditional variance
3. ARCH(q) model
4. GARCH(p, q) model
5. Estimation: maximum likelihood
6. Diagnostics: Engle LM, standardised residuals
7. Volatility forecasting and Value-at-Risk
8. Extensions: EGARCH, GJR-GARCH (overview)

Stylised facts of financial returns

1. Why ARMA is not enough

Standard ARMA models assume a constant variance $\text{Var}(\varepsilon_t) = \sigma^2$. On financial returns $r_t = \log(P_t/P_{t-1})$, this assumption fails:

- the level looks (almost) stationary, but
- large (positive or negative) returns cluster in time;
- small returns also cluster in time;
- the unconditional distribution has heavier tails than $\mathcal{N}(0, \sigma^2)$.

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- small returns also cluster in time;
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Mandelbrot (1963): *“Large changes tend to be followed by large changes, of either sign, and small changes by small changes.”*

Stylised facts (financial returns)

- $\mathbb{E}[r_t] \approx 0$, often a tiny drift.
- Sample ACF of r_t : nearly flat (returns are weakly serially uncorrelated).
- Sample ACF of r_t^2 (or $|r_t|$): *strong, slowly decaying* autocorrelation $\Rightarrow$ volatility clustering.
- Heavy-tailed marginal distribution: kurtosis $\hat{\kappa} > 3$ (excess kurtosis).
- Leverage effect: negative shocks raise volatility more than positive shocks of the same magnitude.

Conditional vs unconditional variance

2. Conditional vs unconditional variance

Information set: $\mathcal{F}_{t-1} = \sigma(r_{t-1}, r_{t-2}, \dots)$.

- *Unconditional* variance: $\sigma^2 = \text{Var}(r_t)$ – the long-run, average risk.
- *Conditional* variance: $\sigma_t^2 = \text{Var}(r_t \mid \mathcal{F}_{t-1})$ – the short-run, predictable risk.

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ARCH/GARCH models specify σ_t^2 as a function of past information, while keeping the unconditional variance σ^2 finite. The series is then *unconditionally homoskedastic* but *conditionally heteroskedastic*.

The two-equation structure

Most volatility models split the dynamics in two:

Mean equation

$$r_t = \mu_t + \varepsilon_t, \quad \varepsilon_t = \sigma_t z_t, \quad z_t \stackrel{\text{i.i.d.}}{\sim} \mathcal{D}(0, 1).$$

μ_t is, e.g., a constant or an ARMA model.

Variance equation

A specification of σ_t^2 in terms of past ε_{t-i}^2 and past σ_{t-j}^2 .

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Common innovation distributions $\mathcal{D}$: standard normal (Gaussian QMLE), Student- t for fat tails, GED.

ARCH(q) model

3. ARCH(q), Engle (1982)

Definition

$$\varepsilon_t = \sigma_t z_t, \quad \sigma_t^2 = \alpha_0 + \sum_{i=1}^q \alpha_i \varepsilon_{t-i}^2,$$

with $z_t \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, 1)$, $\alpha_0 > 0$, $\alpha_i \geq 0$ for $i = 1, \dots, q$.

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- Conditional variance reacts to the *magnitude* of past shocks (squares, sign-blind).
- Robert Engle received the 2003 Nobel Prize in Economics for this model.

Properties of ARCH(q)

Assume $\sum_{i=1}^q \alpha_i < 1$ (covariance-stationarity). Then:

$$\mathbb{E}[\varepsilon_t] = 0, \quad \mathbb{E}[\varepsilon_t^2] = \frac{\alpha_0}{1 - \sum_{i=1}^q \alpha_i} = \sigma^2 \quad (\text{unconditional variance}).$$

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- (ε_t) is white noise (serially uncorrelated) but *not* independent.
- (ε_t^2) follows an AR(q) in disguise: $\varepsilon_t^2 = \alpha_0 + \sum \alpha_i \varepsilon_{t-i}^2 + \nu_t$ with $\nu_t = \varepsilon_t^2 - \sigma_t^2$.
- Excess kurtosis $\hat{\kappa} > 3$: ARCH naturally generates fat tails.

GARCH(p, q) model

4. GARCH(p, q), Bollerslev (1986)

ARCH(q) requires many parameters to capture long memory in volatility. The *generalised ARCH* model adds lagged conditional variances:

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$$\sigma_t^2 = \alpha_0 + \sum_{i=1}^q \alpha_i \varepsilon_{t-i}^2 + \sum_{j=1}^p \beta_j \sigma_{t-j}^2,$$

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Workhorse: GARCH(1, 1) – two parameters (α_1, β_1) on top of α_0 , captures most of the volatility persistence in financial data.

GARCH(1,1): anatomy

$$\sigma_t^2 = \underbrace{\alpha_0}_{\text{baseline}} + \underbrace{\alpha_1 \varepsilon_{t-1}^2}_{\text{news / shock}} + \underbrace{\beta_1 \sigma_{t-1}^2}_{\text{persistence / memory}} .$$

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- α_1 : how strongly today's volatility reacts to yesterday's surprise.
- β_1 : how strongly today's volatility remembers yesterday's volatility.
- $\alpha_1 + \beta_1$ close to 1: very persistent volatility (typical for daily equity returns).

Stationarity and unconditional variance

Covariance-stationarity condition

GARCH(1,1) has a finite unconditional variance if and only if

$$\alpha_1 + \beta_1 < 1.$$

The unconditional variance is then

$$\sigma^2 = \mathbb{E}[\varepsilon_t^2] = \frac{\alpha_0}{1 - \alpha_1 - \beta_1}.$$

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General GARCH(p, q): replace $\alpha_1 + \beta_1$ by $\sum \alpha_i + \sum \beta_j$. **Warning.** If $\alpha_1 + \beta_1 = 1$ we have an *IGARCH* – shocks have permanent effects on volatility.

Estimation

5. MLE for GARCH

Conditional Gaussian density $r_t \mid \mathcal{F}_{t-1} \sim \mathcal{N}(\mu, \sigma_t^2)$:

$$\log \mathcal{L}(\boldsymbol{\theta}) = -\frac{T}{2} \log(2\pi) - \frac{1}{2} \sum_{t=1}^T \left(\log \sigma_t^2(\boldsymbol{\theta}) + \frac{(r_t - \mu)^2}{\sigma_t^2(\boldsymbol{\theta})} \right).$$

- σ_t^2 is computed recursively from the model equation.
- Initialised with $\sigma_1^2 = \hat{\sigma}^2$ (sample variance) by convention.
- Maximisation by quasi-Newton; the Gaussian QMLE is consistent and asymptotically normal even if the true z_t is non-Gaussian (heavy-tailed).

Estimation – Python (arch package)

```
from arch import arch_model

am = arch_model(returns * 100,          # rescale for numerics
                mean="Constant",
                vol="GARCH",
                p=1, q=1, dist="normal")
res = am.fit(displ="off")
print(res.summary())
print("alpha + beta =",
      res.params["alpha[1]"] + res.params["beta[1]"])
```

- vol="ARCH" for pure ARCH(q).
- dist="t" for Student- t innovations.
- res.conditional_volatility stores $\hat{\sigma}_t$.

Diagnostics

6. Engle's LM test for ARCH effects

Run the auxiliary regression

$$\hat{\varepsilon}_t^2 = \gamma_0 + \gamma_1 \hat{\varepsilon}_{t-1}^2 + \cdots + \gamma_q \hat{\varepsilon}_{t-q}^2 + u_t.$$

- $H_0: \gamma_1 = \cdots = \gamma_q = 0$ (no ARCH effect).
- Test statistic: $LM = T \cdot R^2$, with T the sample size and R^2 the regression's coefficient of determination.
- Under H_0 , $LM \xrightarrow{d} \chi_q^2$.
- Reject H_0 if $LM > \chi_{q,1-\alpha}^2$ – evidence for ARCH effects, fit a GARCH model.

After fitting GARCH, check

$$\hat{z}_t = \frac{\hat{\varepsilon}_t}{\hat{\sigma}_t}.$$

A well-fitted model satisfies:

- Sample ACF of $\hat{z}_t$ inside Bartlett bands.
- Sample ACF of $\hat{z}_t^2$ also inside the bands (volatility clustering removed).
- Engle LM on $\hat{z}_t^2$ does *not* reject.
- Marginal distribution closer to the assumed $\mathcal{D}(0, 1)$ (QQ-plot).

Forecasting and Value-at-Risk

7. One-step-ahead variance forecast

For GARCH(1,1),

$$\hat{\sigma}_{T+1}^2 = \alpha_0 + \alpha_1 \hat{\varepsilon}_T^2 + \beta_1 \hat{\sigma}_T^2.$$

Multi-step iterated forecast (mean-reverting to the unconditional variance):

$$\hat{\sigma}_{T+h}^2 = \sigma^2 + (\alpha_1 + \beta_1)^{h-1} (\hat{\sigma}_{T+1}^2 - \sigma^2).$$

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$\hat{\sigma}_{T+h}^2 \rightarrow \sigma^2$ as $h \rightarrow \infty$, at a rate controlled by the persistence $\alpha_1 + \beta_1$.

At horizon 1, level α (e.g. 1%), assuming Gaussian innovations:

$$\text{VaR}_{T+1}(\alpha) = -(\hat{\mu} + z_{\alpha} \hat{\sigma}_{T+1}),$$

with $z_{\alpha} = \Phi^{-1}(\alpha)$ – e.g. $z_{0.01} = -2.326$, $z_{0.05} = -1.645$.

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- Static (unconditional) VaR uses sample σ instead of $\hat{\sigma}_{T+1}$.
- GARCH-VaR is responsive to the current volatility regime and produces tighter intervals in calm periods, wider ones in turbulent periods.

Extensions (overview)

8. Extensions

EGARCH (Nelson, 1991)

Models $\log \sigma_t^2$ to relax non-negativity constraints and capture asymmetric responses (leverage effect):

$$\log \sigma_t^2 = \omega + \alpha g(z_{t-1}) + \beta \log \sigma_{t-1}^2.$$

GJR-GARCH (Glosten–Jagannathan–Runkle, 1993)

Adds a threshold term that activates only on negative shocks:

$$\sigma_t^2 = \omega + (\alpha + \gamma \mathbb{1}_{\varepsilon_{t-1} < 0}) \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2.$$

$\gamma > 0$: bad news raises volatility more than good news.

Take-home messages

- Returns look uncorrelated, but their squares are *strongly* autocorrelated – volatility clustering.
- ARCH(q) models conditional variance as an AR on past squared shocks; GARCH(p, q) adds memory via past variances.
- GARCH(1, 1) is the standard workhorse: $\alpha_1 + \beta_1 < 1$ for stationarity, with unconditional variance $\alpha_0/(1 - \alpha_1 - \beta_1)$.
- Test for ARCH effects with Engle's LM; estimate by MLE (arch package); validate with standardised residuals.
- Conditional VaR uses $\hat{\sigma}_{T+1}$ instead of the sample $\hat{\sigma}$ – it adapts to the current regime.